

Erdős Problem #87

EP-87

L1 - TRACTABLE

OPEN

Graph Theory

ORIGINAL PROBLEM STATEMENT

Let $\epsilon > 0$. Is it true that, if k is sufficiently large, then $R(G) > (1 - \epsilon)^k R(k)$ for every graph G with chromatic number $\chi(G) = k$?

Even stronger, is there some $c > 0$ such that, for all large k , $R(G) > cR(k)$ for every graph G with chromatic number $\chi(G) = k$?

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsignedMath>